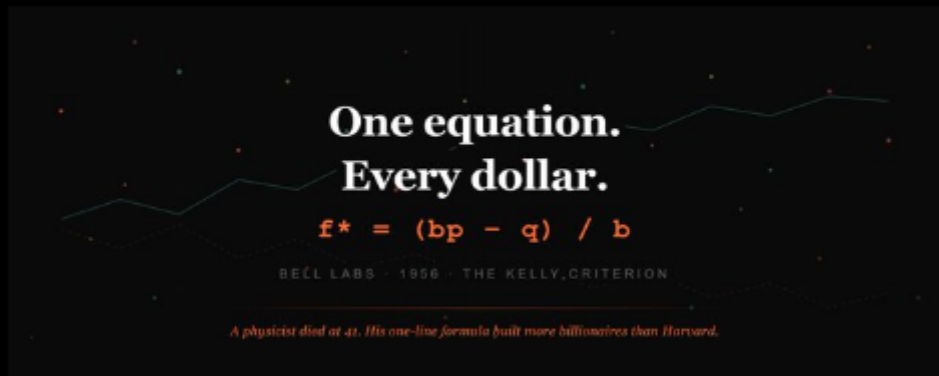



Author

Tigerflow

@tigerflow

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One Equation Controls Every Dollar You Will Ever Make in Markets. You Have Never Seen It

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One Equation Controls Every Dollar You Will Ever Make in Markets. You Have Never Seen It.

A PHYSICIST DIED AT 41. HIS ONE-LINE FORMULA BUILT MORE BILLIONAIRES THAN HARVARD.

In 1956, a physicist at Bell Labs published a ten-page paper in a journal nobody outside the building read. The paper solved a problem that had no name at the time: **given a real edge, how much of your capital should you bet?**

The answer was a single formula. It was so simple it fit in one line. It was

The answer was a single formula. It was so simple it fit in one line. It was so powerful that within five years, a mathematics professor used it to beat every casino in Las Vegas, then turned it on the stock market and ran a fund that returned 20% a year for 30 years without a single down year.

The physicist's name was John Larry Kelly Jr. He died in 1965 at age 41 of a brain hemorrhage on a Manhattan sidewalk. He never traded a share in his life. **The formula he left behind has generated more wealth than any other equation in the history of finance.** Almost nobody who trades has heard of it.

Here is the equation, the proof, the trap, and the reason every account you have ever blown up was a sizing problem, not a strategy problem.

THE KELLY CRITERION · BELL LABS, 1956

The entire formula.

$$f^* = \frac{p - q}{b}$$

f^* = fraction of capital to risk · p = probability of winning · q = probability of losing ($1 - p$)
· b = ratio of win to loss

Layer 1 - The Sizing Illusion

You do not have a strategy problem. You have a sizing problem.

Ask any trader what blew up their account and they will tell you a story about a bad trade. A news event. A stop that didn't trigger. A position

about a bad trade. A news event. A stop that didn't trigger. A position they held too long.

They are almost always wrong. The trade was not the problem. **The size was.**

Take two traders. Identical strategy, identical signals, identical win rate of 55%, identical 2:1 reward-to-risk. Trader A risks 2% per trade. Trader B risks 25% per trade. Run them both for a hundred trades. The result is not a difference in degree. It is a difference in kind: one is rich, one is bankrupt. Same strategy. Same market. Same year.

SAME STRATEGY. DIFFERENT SIZING. 100 TRADES AT 55% WIN RATE, 2:1 R:R.

METRIC	TRADER A (2%)	TRADER B (25%)
Starting capital	\$10,000	\$10,000
Ending capital	\$14,890	\$0.03
Max drawdown	-12%	-99.9%
Longest losing streak survived	9 in a row	4 before ruin
Probability of ruin	~0%	~100%

Trader B did nothing wrong except size. That is the illusion. The strategy was never the variable. The bet size was always the only thing that mattered.

"It's not whether you're right or wrong that's important, but how much money you make when you're right and how much you lose when you're wrong." - **George Soros**

Layer 2 - The Proof That Optimal Exists

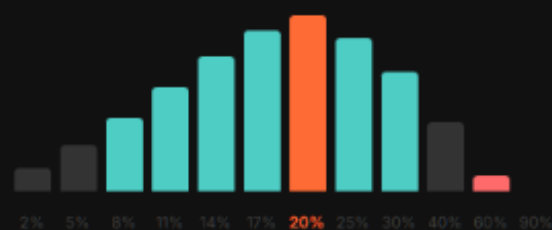
Kelly proved it is not a matter of opinion. There is a mathematically correct answer.

Before Kelly, position sizing was vibes. Gut feeling. "Risk what you can afford to lose." This is the financial equivalent of a doctor saying "eat healthy."

Kelly showed something nobody expected: **for any edge, there is exactly one bet size that maximizes the long-run growth rate of your capital.** Bet less, and you leave money on the table. Bet more, and you destroy yourself. The function is not linear. It is a curve with a sharp peak and a brutal cliff.

The Kelly Curve.

GROWTH RATE VS. FRACTION OF CAPITAL RISKED



Peak = Kelly optimal · Right of peak = ruin approaches exponentially

The right side of that curve is where every blown account lives. You were to the right of Kelly. That is the entire diagnosis.

Ed Thorp, the MIT professor who took Kelly's formula from Bell Labs to blackjack tables to Wall Street, ran it for three decades. His fund, Princeton Newport Partners, returned 20% annually for 30 years. He said the same thing Kelly's math said, just in English:

"The signal was never the edge. The sizing was."-Ed Thorp

Layer 3 - The Code

Twelve lines. That is all it takes.

Here is the Kelly Criterion as working code. Plug in your win rate and your reward-to-risk ratio. It returns exactly how much of your capital to allocate to the next trade. This is the formula that ran inside Ed Thorp's fund, inside Renaissance Technologies, inside every serious quantitative shop on the planet.

python

```
def kelly_fraction(win_rate, reward_risk_ratio):  
    """  
    Returns optimal fraction of capital to risk.  
    win_rate: probability of a winning trade (0 to 1)  
    reward_risk_ratio: avg_win / avg_loss  
    """  
    p = win_rate  
    q = 1 - p  
    b = reward_risk_ratio
```

python



```
def kelly_fraction(win_rate, reward_risk_ratio):
    """
    Returns optimal fraction of capital to risk.
    win_rate: probability of a winning trade (0 to 1)
    reward_risk_ratio: avg_win / avg_loss
    """
    p = win_rate
    q = 1 - p
    b = reward_risk_ratio

    f = (p * b - q) / b

    return max(0, f) # never negative

# — Example —
win_rate      = 0.55 # 55% win rate
reward_risk   = 2.0  # make $2 for every $1 risked

kelly = kelly_fraction(win_rate, reward_risk)
print(f"Optimal fraction: {kelly:.1%}")
print(f"Half-Kelly (safer): {kelly/2:.1%}")

# Output:
# Optimal fraction: 32.5%
# Half-Kelly (safer): 16.2%
```

Notice the output. A 55% win rate with 2:1 R:R yields a Kelly fraction of 32.5%. That is the mathematical maximum. In practice, no serious fund uses full Kelly. The volatility is unbearable. Most use half-Kelly or less. **Thorp used half. Buffett, without knowing the name, uses roughly a quarter.**

Layer 4 – The Trap Nobody Talks About

The formula is perfect. Your inputs are not.

Here is where most people destroy themselves with Kelly. The formula assumes you know your true win rate and your true reward-to-risk. You do not.

assumes you know your true win rate and your true reward-to-risk. You do not.

You have a backtest. A sample of fifty trades. Maybe a hundred. You think your win rate is 60%. In reality, your true win rate could be anywhere from 45% to 72%. Kelly does not care about your estimate. It runs on the real number. If you plug in 60% and the truth is 48%, full Kelly will annihilate you faster than no system at all.

KELLY SIZING AT DIFFERENT ESTIMATION ERRORS · STARTING \$100K, 200 TRADES

ESTIMATED WIN RATE	TRUE WIN RATE	KELLY SIZE	RESULT AFTER 200 TRADES
60%	60%	20%	\$847,000
60%	55%	20%	\$112,000
60%	50%	20%	\$13,400
60%	45%	20%	\$890
60% (half-Kelly)	45%	10%	\$41,200

Look at the last row. Half-Kelly with a **badly wrong estimate** still leaves you alive. Full Kelly with the same error leaves you with lunch money. This is why Thorp, the man who literally proved Kelly works, told everyone the same thing for thirty years:

"Use half Kelly. Full Kelly is for God, who knows the true probabilities." - Ed Thorp

Layer 5 - The Compounding Secret Hidden in the Math

Layer 5 - The Compounding Secret Hidden in the Math

Kelly does not maximize profit. It maximizes the growth rate. The difference is everything.

Most traders optimize for the biggest possible win. Kelly optimizes for something completely different: **the speed at which your capital multiplies over time.**

This distinction is subtle and deadly. A strategy that maximizes expected profit can still go to zero. A strategy that maximizes geometric growth rate cannot. Kelly found the only sizing rule in all of mathematics that guarantees you will outperform every other fixed-fraction strategy over time, with probability one.

The proof is elegant. Over N bets, your capital is:

```
mathematica
# After N trades, capital grows as:


$$C(N) = C_0 \times (1 + f \cdot b)^W \times (1 - f)^L$$


# Where:
# C0 = starting capital
# f = fraction risked per trade
# b = reward/risk ratio
# W = number of wins
# L = number of losses

# Kelly maximizes  $\log(C(N)) / N$ 
# → the geometric growth rate
# → the only rate that compounds
```

This is why Kelly beats everything in the long run. Not because it wins more. Because it compounds faster. And compounding is the only force in markets that turns a small edge into a fortune.

"The big money is not in the buying and selling, but in the waiting." - **Charlie Munger**

Layer 6 - The Reality Layer

The map is not the territory. Here is how to actually use it.

Kelly in its pure form assumes one bet at a time, known probabilities, and no correlation between bets. Markets violate all three. So the practitioners who have used it successfully for decades all apply the same set of adjustments:

Use fractional Kelly. Half-Kelly cuts your growth rate by only 25% but cuts your variance by 50%. The trade is overwhelmingly worth it. Quarter-Kelly is even safer and still compounds respectably.

Update your inputs constantly. Your win rate is not a fixed number. It drifts. It decays. If you measured it six months ago, it is already wrong. The edge you have this year decays because the world updates on your presence.

Cap your maximum position. Even if Kelly says 40%, never risk more than 5-6% on a single trade. The estimation error alone justifies the cap. Andrew Lo at MIT calls this "the Kelly gap" - the distance between theoretical optimal and practical optimal is where all the adult money lives.

PRACTICAL KELLY IMPLEMENTATION GUIDE

PARAMETER	THEORETICAL	PRACTICAL (WHAT FUNDS ACTUALLY DO)
Kelly fraction	Full Kelly	¼ to ½ Kelly

The Stack

Six layers. One equation. All of it free since 1956.

1. The size is the strategy. Everything else is noise. – Kelly, 1956
 2. There is a mathematically optimal bet. It is a curve, not a guess. – Shannon, Thorp
 3. The formula is twelve lines of code. You can run it tonight. – Implementation
 4. Your inputs are wrong. Use half-Kelly. Stay alive. – Thorp, 30 years of proof
 5. Kelly maximizes growth rate, not profit. Compounding is the edge. – Bernoulli, 1738
 6. The map is not the territory. Cap, update, and fraction. – Lo, Derman
-

The Close

Every blown account in history shares one feature. The trader was to the right of the Kelly curve. They did not have a bad strategy. They had a lethal size. The formula that corrects this has been sitting in a Bell Labs paper since Eisenhower was president.

Twenty people built the mathematics. One equation distilled it. A handful of funds took it seriously and compounded for decades. The rest of the industry spent seventy years arguing about entries and exits while the only variable that mattered – how much – sat in the library, free, waiting.

You do not need a better signal. You need a better size.

The edge was never the entry.
The edge is *the size*.

This took 40+ hours to research, write, and build. If it saved you from even one blown account, it paid for itself.

Follow @tigerflow for the next deep dive. I break down the math, the history, and the frameworks that separate the 1% from the graveyard - one layer at a time. No fluff. No signals. Just the stack.

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